Hackathon

Development of AI-Powered, Interoperable, and Automated Solutions for Hospitals and MedTech.

Jointly hosted by
Ratan Tata Innovation Hub (RTIH)

Background & Context

Hospitals and healthcare providers face mounting challenges in patient safety, diagnostic efficiency, compliance monitoring, and device interoperability. Manual monitoring and fragmented workflows often delay critical interventions, increasing risks for patients and burdening staff. Meanwhile, medical devices and diagnostic equipment frequently lack seamless integration with hospital IT systems, limiting real-time data use and slowing clinical decision-making.

To address these gaps, there is a pressing need for affordable, intelligent, and interoperable healthcare automation solutions. By leveraging AI, automation, and smart device integration, hospitals can improve patient outcomes, streamline workflows, and strengthen compliance with accreditation standards.

Challenge Overview

The Hackathon invites startups, innovators, institutions, and technology providers to design and develop next-generation solutions that:

- Enhance patient monitoring and safety through AI-powered systems.
- Streamline diagnostic workflows with automation and AI-enabled reporting.
- Strengthen compliance monitoring with integrated dashboards.
- Enable interoperability of medical devices with hospital IT systems.
- Deploy fully automated diagnostic platforms for faster, more accurate results.

Solutions should be scalable, secure, and adaptable for deployment across hospitals, diagnostic centers, and MedTech ecosystems.

I. Challenge Vision and Rationale

Overarching Goal

To develop intelligent, interoperable, and automated healthcare solutions that improve patient safety, diagnostic efficiency, and compliance outcomes while reducing staff workload.

Rationale

Current hospital systems rely heavily on manual monitoring, fragmented workflows, and non-integrated devices, which delay interventions and increase risks. AI-powered automation and interoperability can bridge these gaps, enabling predictive alerts, streamlined diagnostics, and real-time data sharing. This challenge seeks to foster innovation in healthcare automation, resulting in solutions that are practical, secure, and scalable for real-world adoption.

Core Objectives

1. Enhance patient monitoring and safety through AI-powered systems.
2. Streamline diagnostic workflows with automation and AI-enabled reporting.
3. Strengthen compliance monitoring with integrated dashboards.
4. Enable interoperability of medical devices with hospital IT systems.
5. Deploy fully automated diagnostic platforms for faster, more accurate results.
6. Solutions should be scalable, secure, and adaptable for deployment across hospitals, diagnostic centers, and MedTech ecosystems

II. Targeted Participants and Eligibility

The challenge is open to participants across the construction and robotics ecosystem, including:

Category	Description
Startups (Early to Growth Stage)	Healthcare and MedTech startups developing AI-powered or automated solutions
Independent Innovators	Individual developers or innovators with functional healthcare prototypes
Academic & Research Institutions	Universities, research labs, and incubators working on healthcare technologies
Technology Companies	Firms with existing healthcare or MedTech solutions ready for customization

III. Scope of the Solution

Participants are expected to design solutions covering any of the following functional scope:

Core Functional Requirements (Indicative areas: Non-Exhaustive)

Hospitals

- AI-powered Early Warning Systems (EWS) for continuous patient monitoring.
- Automated diagnostics with AI-enabled reporting.
- Compliance dashboards integrated with patient monitoring and diagnostics.

MedTech

- Interoperable smart devices with automated data sharing and AI analytics.
- Fully automated diagnostic platforms with AI-driven analysis.

Platform Quality

- Secure and compliant with healthcare standards.
- Scalable and modular architecture.
- User-friendly design for clinicians and staff.

Participants are free to suggest focus areas beyond the above but aligned to the core objectives.

IV. Evaluation and Selection Criteria

Proposals will be evaluated by an expert committee based on the following criteria:

Evaluation Parameter	**Weightage**
Innovation & Originality	20
Technical Completeness & Scalability	10
User Experience	20
Real-World Applicability & Cost Effectiveness	40
Documentation & Demo Quality	10
Total	**100**

V. Rewards & Adoption Pathway

VI. Expected Deliverables

Participants must submit:

- A working prototype of healthcare automation solution(s)
- Source code repository with setup documentation
- Use-case demonstrations for hospitals and MedTech applications
- System architecture diagram
- Demo video or live presentation

VII. Goal & Expected Impact

Aims to create a robust, inclusive, and sustainable healthcare ecosystem that:

- Enhances patient safety and monitoring.
- Streamlines diagnostic workflows and reporting.
- Strengthens compliance and accreditation outcomes.
- Improves interoperability of medical devices.
- Supports real-time clinical decision-making.